Sample Solutions Jan 2023 - Statistics & Probability Questions

Q1

e) 12, 10, 18, 14, 8, 10

i) mean $\bar{x} = \frac{\sum x_i}{n} = \frac{12+10+18+14+8+10}{6} = \frac{72}{6} = \boxed{12}$

ii) stdev $s^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n-1}$

$= \frac{12^2+10^2+18^2+14^2+8^2+10^2 - 6(12)^2}{6-1}$

$= \frac{928 - 864}{5}$

$= \frac{64}{5}$

$s^2 = 12.8$

$s = \sqrt{12.8} = \boxed{3.58}$

f) two fair 6-sided dice - 2 dice so total possibilities is 36.

i) P[1 and 4] = $2/36 = \boxed{1/18}$

(1,4) or (4,1) - 2 possible out of 36.

$\frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} = 2/36$.

ii)

SUM	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

P[SUM ≥ 3] = $35/36$.

Note: ensure you build the correct event space. SUM - addition OR PRODUCT - Multiplication depending on what Q asks.

Question 3.

a) computers in eight rooms 3, 15, 1, 2, 1, 30, 2, 1

i) median: sort in ascending order

1, 1, 1, 2 | 2, 3, 15, 30

Median

$\frac{2+2}{2} = \frac{4}{2} = \boxed{2}$

ii) mode - most frequent item = $\boxed{1}$

iii) most sensible measurement for average of this dataset: **median**

When there are outliers/extreme values (i.e. 15, 30) median is a more appropriate measure to use as mean value ($\bar{x} = 6.9$) is affected by the outliers. i.e. very large extreme values pull the mean up & very small extreme values pull the mean down. Median is the middle value (point) in the dataset & so is not affected by outliers or extreme values.

iv) Need min, max values, 1st Quartile, 3rd Quartile and median values to build the boxplot. (Note: not taking outliers (1.5 times interquartile range into account in the diagram here)

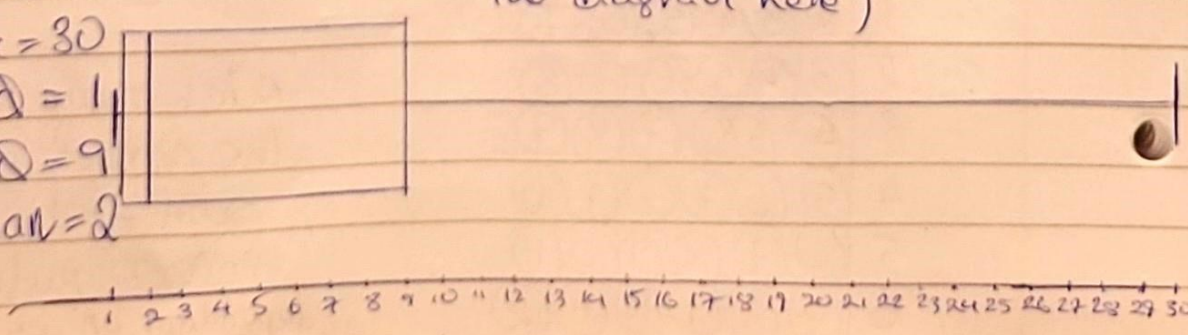
Min = 1

Max = 30

1st Q = 1

3rd Q = 9

Median = 2



b)

10. Books	Freq	m_i	$f_i m_i$	m_i^2	$f_i m_i^2$
0 - 4	2	2	2×2 = 4	4	2×4 = 8
5 - 9	4	7	4×7 = 28	49	4×49 = 196
10 - 14	6	12	6×12 = 72	144	6×144 = 864
15 - 19	8	17	8×17 = 136	289	8×289 = 2312
	20		240		3380

i) Mean (Grouped Data) : $\bar{x} = \frac{\Sigma f_i m_i}{\Sigma f_i}$ — midpoint

$= \frac{4+28+72+136}{20}$

$= \frac{240}{20}$

$\bar{x} = \boxed{12}$

ii) stdev (Grouped Data) : $s^2 = \frac{\Sigma f_i m_i^2 - M(\bar{x})^2}{M-1}$ — total freq Σf_i

$= \frac{8+196+864+2312 - 20(12)^2}{20-1}$

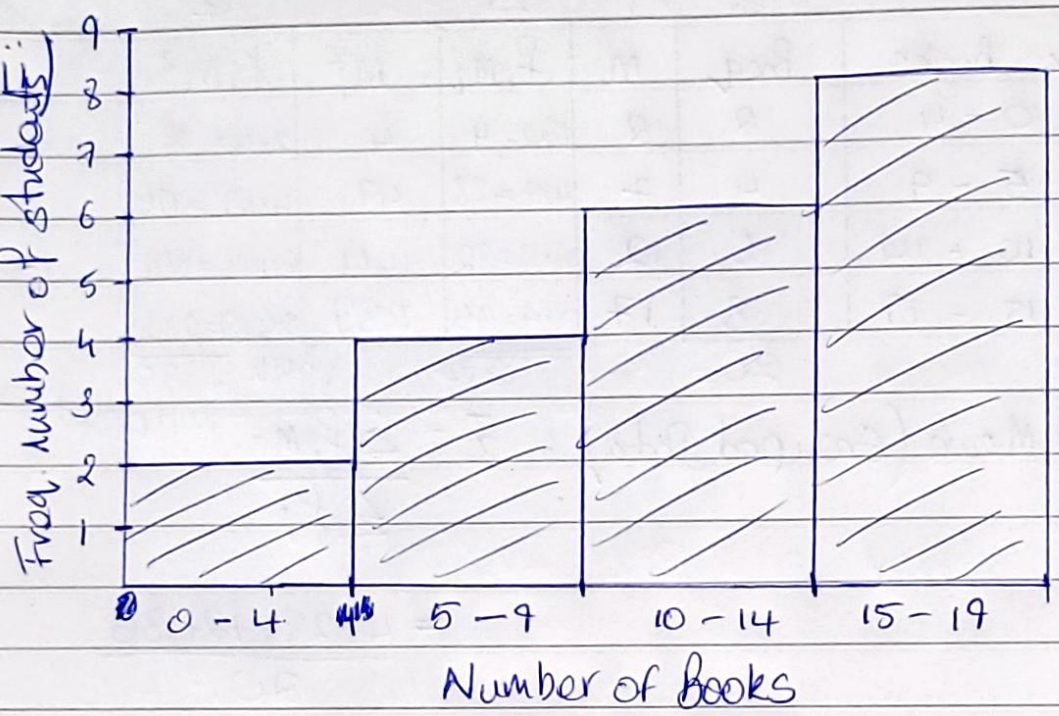
$= \frac{3380 - 2880}{19}$

$= \frac{500}{19}$

$s^2 = 26.32$

$s = \sqrt{26.32}$

$= \boxed{5.1}$



c) average 3 out of 5 customers.
random sample of 10
$P[X=6]$ Binomial distribution

$$P[x=r] = {}^{n}C_r\, p^r q^{n-r} = \frac{n!}{(n-r)!\,r!}\, p^r q^{n-r}$$

$n = 10$
$r = 6$
$p = 3/5 = 0.6$
$q = 1 - p = 0.4$

$$P[X=6] = \frac{10!}{(10-6)!\,6!} \times 0.6^{6} \times 0.4^{10-6}$$

$$= \boxed{0.25082}.$$